

Use NobootDeviceChecks tool to troubleshoot Bugcheck 7B_RDP SSH

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Tags

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When to use the tool:

The tool is mainly designed for help with troubleshooting/isolating No boot issues with bugcheck 7B (Thank you Lucia Chen for creating the tool). 0X7B means INACCESSIBLE BOOT Device. It is a Windows crash issue which happens when Windows tries to start, but fails to create/access the OS Disk/Volume or is unable to load any of the boot drivers for some reason (missing or corrupt driver, driver disabled in registry, etc.).

However, what makes 0x7B difficult to troubleshoot is:

1. Because of 0x7B, Windows fails to boot, we can only check the system in offline mode. Unlike online mode, in which we can check what the failed device is and what drivers it requires, in offline mode, what we have is only registry hive file (SYSTEM) and system files. We need to understand structure of Windows device registry and driver store to understand what device is failing and what configuration/driver the device is using.
2. Although it is a 0x7B bugcheck/crash, there is often no dump file (this is an interesting paradox: you need to write dump data to the disk upon crash, but currently, this kind of crash means it is inaccessible.).

How to use it?

Here is how it works:

1. Create a rescue VM environment by following steps from the article <https://learn.microsoft.com/en-us/troubleshoot/azure/virtual-machines/troubleshoot-vm-by-use-nested-virtualization> and install Hyper V role on the rescue VM.
2. Download the NoBootDeviceCheck exe from here and share with the customer [NoBootDeviceChecks](#) . Please ask the customer to copy the file to the Rescue VM.
3. Attach the affected VM's OS disk as a data disk on the Rescue VM.
4. Launch CMD as an admin on the Rescue VM, navigate to the location where NoBootDeviceCheck exe is saved, and run the tool against the OS drive of the affected VM:

```
nobootdevicecheck.exe H:
```



(Disk 3 partition 2)	Simple	Basic		Healthy (P...	120.5
System (G:)	Simple	Basic	NTFS	Healthy (A...	500 M
System Reserved	Simple	Basic	NTFS	Healthy (S...	500 M
Temporary Storag...	Simple	Basic	NTFS	Healthy (P...	16.00
Windows (C:)	Simple	Basic	NTFS	Healthy (B...	126.51
Windows (H:)	Simple	Basic	NTFS	Healthy (P...	126.51

Disk 2		
Basic		
127.00 GB		
Online		
	System (G:)	Windows (H:)
	500 MB NTFS	126.51 GB NTFS
	Healthy (Active, Primary Partition)	Healthy (Primary Partition)
Disk 3		
Basic		
127.00 GB		
Offline		
	500 MB	126.51 GB
	Healthy (Active, Primary Partition)	Healthy (Primary Partition)

```

Administrator: Command Prompt
C:\Users\Manpreet\Desktop>NOBOOTDEVICECHECK.EXE H:
Boot Disk Device Tree Checker
#####
INFO: Online Device Tree Learning Started.
*****
INFO: Online Device Tree Learning was Successful.
*****
INFO: Online Device Tree Learning Ended.

INFO: Showing Volume Device Tree:
|----Volume Manager
|----Volume

INFO: Showing Disk Device Tree:
|----ACPI x64-based PC
|----Microsoft ACPI-Compliant System
|----PCI Bus
|----PCI to ISA Bridge
|----Microsoft Hyper-V Virtual Machine Bus
|----Microsoft Hyper-V SCSI Controller
|----Disk drive

INFO: Offline Check Started on Volume Device Node.
*****

```

5. The tool will show online device tree of OS disk and volume device and also, offline check result. "Alert" messages are the ones you need to pay attention to.

The output can be saved into a txt file by adding ">output.txt" at the end the command.

Running the tool without any parameter will show help:

```
#####
We need a drive letter as input, like 'c:'.
```

This tool does two operations:

1. Scan the input driver letter related disk and volume device, learn the device tree of its depending devices.
2. Check registry and files to verify if all the device related registry entries and files are installed properly.

How to understand the result and take next actions

The first part in output is the device tree. A sample output:

```
=====
INFO: Showing Volume Device Tree:
|----Volume Manager
|----Volume
INFO: Showing Disk Device Tree:
|----ACPI x64-based PC
|----Microsoft ACPI-Compliant System
|----PCI Bus
|----Intel(R) 82371AB/EB PCI Bus Master IDE Controller
|----IDE Channel
|----Microsoft Hyper-V Accelerated Disk Drive
.=====
```

Those device trees can help us to understand clearly what devices are related to the OS disk and volume device. All the devices on the tree must running properly in order to have working disk and volume device.

The second part is the offline scan result: =====

```
INFO: Offline Check on Device: Microsoft ACPI-Compliant System
INFO:Checking on Device Driver Service:
INFO:Checking on Driver File ACPI...
Alert: Start value for this driver is NOT BOOT!
Alert: Driver File is not located in Offline system, the Error is: The system cannot find the file specified.
INFO:Checking on Device Filter Drivers:
INFO:Checking on Device Driver Service configured in offline system:
INFO:Checking on Driver File ACPI...
Alert: Start value for this driver is NOT BOOT!
Alert: Driver File is not located in Offline system, the Error is: The system cannot find the file specified.
INFO:Checking on Device Filter Drivers configured in offline system:
INFO: Offline Check on this Device ended with some issue detected.
```

=====

The scan is done on every device on the devices trees above. The scan is to make sure all related drivers are enabled, having properly registry entries and files.

If any issue is detected, a "Alert" message will be print out. The sample above shows volmgr driver is detected missing in drivers folder on offline system.

Next Actions

Issues detected in "Alert" messages tell us the direct cause: what device and what driver is having issue. The next step is to check why the driver/device shows up in the alert message. Normally, it is if either driver file/entry is missing, or driver is not enabled. This could be a result of improper driver package uninstallation, software configuration, etc.

The sample output above shows ACPI driver is not set as BOOT start and also, the file is missing under drivers folder.

We can copy the file over from OS Drive > Windows > System32 > DriverStore > FileRepository folder of the affected VM, or from an identical server with same OS and similar patch level.

If the driver is disabled, we can enable it in Registry by loading the affected VM's registry hive, navigating to Registry path HKLM > System > Controlset00X > Services > driver key, and changing the Start value of the driver to 0.

Additional Information:

[How to use 7Bchecks.iso](#)

[Advanced Troubleshooting for inaccessible Boot device](#)

Escalation

1. If this doesn't work out, please reach out to the [Unable to RDP-SSH SME channel on teams](#) for advise providing the case number, issue description and your question

Need additional help or have feedback?

<i>To engage the Azure RDP-SSH SMEs...</i>	<i>To provide feedback on this page...</i>	<i>To provide kudos on this page...</i>
If you are in the OneVM team then reach out to the RDP-SSH SMEs AVA channel via Teams. If you are a VM Delivery Partner then follow your team's process for swarming or ask your TA. Make sure to use the Ava process for faster assistance.	Use the RDP-SSH Feedback form to submit detailed feedback on improvements or new content ideas for RDP-SSH. Please note the link to the page is required when submitting feedback on existing pages! If it is a new content idea, please put N/A in the Wiki Page Link.	Use the RDP-SSH Kudos form to submit kudos on the page. Kudos will help us improve our wiki content overall! Please note the link to the page is required when submitting kudos!